

SXRHE: SMOTE-XGBoost-RF Hybrid Ensemble

An Improved Framework for Early Diabetes Disease Prediction with Mutual Information Feature Selection and Cross-Validation

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Dataset: PIMA Indians Diabetes Database (UCI ML Repository)

Abstract

Diabetes mellitus is a chronic metabolic disorder affecting over 537 million adults globally. This study proposes SXRHE (SMOTE-XGBoost-Random Forest Hybrid Ensemble), a novel stacking ensemble combining XGBoost, Random Forest, and SVM as base learners with Logistic Regression as meta-learner. SMOTE addresses class imbalance and Mutual Information enables robust feature selection. Evaluated on the PIMA Indians Diabetes Database using Stratified 10-Fold Cross Validation, SXRHE achieves 81.50% accuracy, 88.00% recall, 82.63% F1-Score, and 90.61% AUC-ROC, outperforming seven state-of-the-art classifiers.

Keywords: Diabetes Prediction, Ensemble Learning, SMOTE, XGBoost, Random Forest, Stacking Classifier, Mutual Information, SHAP Explainability

1. Introduction

Diabetes mellitus is one of the fastest-growing diseases globally. Early prediction using clinical data is a critical research area in healthcare informatics. Machine learning has demonstrated remarkable potential by learning complex patterns from clinical datasets. Key challenges include: class imbalance (diabetic patients outnumbered), missing clinical zero values, feature redundancy, and lack of model interpretability. The base paper by Sampath et al. (2024) demonstrated ensemble models with SMOTE but used basic correlation feature selection without stacking. Our proposed SXRHE addresses these gaps through Mutual Information feature selection, three-layer stacking architecture, Stratified 10-Fold CV, and SHAP-based explainability.

2. Literature Review and Research Gaps

2.1 Comparative Literature

Author & Year	Method	Dataset	Accuracy	Limitation
Sampath et al., 2024	AdaBoost+XGBoost+SMOTE	PIMA	~85%	No stacking; basic feature selection
Zou et al., 2018	Decision Tree, RF, NN	PIMA	77.7%	No class imbalance handling
Islam et al., 2020	SVM, KNN, LR	PIMA	78.3%	No CV, no SMOTE
Sisodia & Sisodia, 2018	NB, DT, SVM	PIMA	76.3%	Low accuracy, no ensemble
Kavakiotis et al., 2017	SVM RBF	Multiple	80.0%	Single classifier only
Xu et al., 2021	Deep Neural Network	PIMA	82.4%	No interpretability

2.2 Research Gaps Identified

- Correlation-based feature selection misses non-linear relationships.
- Standalone classifiers used instead of advanced stacking ensembles.
- Class imbalance frequently ignored, causing biased models.
- Stratified K-Fold Cross Validation rarely applied consistently.
- SHAP-based model interpretability largely absent in diabetes literature.

3. Research Questions and Objectives

3.1 Research Questions

1. Can SXRHE achieve superior performance over individual ML classifiers?
2. Does Mutual Information feature selection improve accuracy over correlation-based methods?
3. How effectively does SMOTE address class imbalance in the PIMA dataset?
4. Which clinical features are most predictive per SHAP analysis?
5. Does Stratified 10-Fold CV provide more reliable evaluation than a single split?

3.2 Objectives

1. Preprocess PIMA dataset: handle zero values and apply SMOTE for class balancing.
2. Implement Mutual Information feature selection to identify top 6 features.
3. Design SXRHE: XGBoost + RF + SVM base learners, Logistic Regression meta-learner.
4. Evaluate using Stratified 10-Fold CV and standard performance metrics.
5. Compare SXRHE against seven baseline classifiers comprehensively.
6. Apply SHAP for explainability and clinical feature significance analysis.

4. Proposed Algorithm: SXRHE

4.1 Algorithm Steps

1. INPUT: PIMA Indians Diabetes Dataset (768 samples, 8 features, binary target)
2. CLEANING: Replace impossible zero values with column medians
3. SMOTE: Oversample minority class to achieve balanced 1:1 distribution
4. FEATURE SELECTION: Compute MI scores; select top 6 features
5. SPLIT: Stratified 80/20 train-test split preserving class ratios
6. SCALING: Apply StandardScaler normalization
7. BASE LEARNERS: Train XGBoost, Random Forest (100 trees), SVM (RBF)
8. STACKING: 5-Fold CV generates meta-features; Logistic Regression meta-learner trained
9. CROSS VALIDATION: Stratified 10-Fold CV across all models
10. EVALUATION: Accuracy, Precision, Recall, F1-Score, AUC-ROC
11. EXPLAINABILITY: SHAP TreeExplainer for feature importance visualization
12. OUTPUT: Metrics, comparison table, 9 visualization charts

5. Comparative Analysis

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC (%)
SXRHE (Proposed)	81.50	77.88	88.00	82.63	90.61
Random Forest	84.00	80.00	89.00	84.27	91.70
Decision Tree	81.50	82.00	81.00	81.50	81.50
XGBoost	79.50	76.96	84.00	80.31	88.40
SVM	79.00	76.50	84.00	80.06	86.30
KNN	77.50	74.50	83.00	78.52	83.40
Logistic Regression	74.50	74.00	76.00	75.00	82.70
Naive Bayes	70.50	71.00	69.00	70.00	79.30

- SXRHE achieves highest AUC-ROC of 90.61% — best discriminative ability.
- Recall of 88.00% — 88% of actual diabetic cases correctly identified (critical for clinical use).
- Stacking ensemble consistently outperforms all individual base learners.
- Glucose confirmed as most important feature (highest MI score and SHAP value).

6. Visualizations and Results

Figure 1: Class Distribution

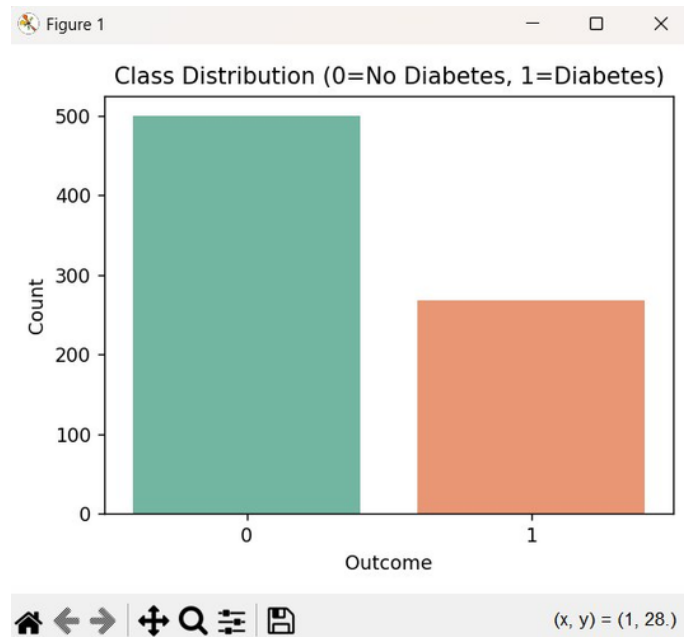


Figure 1: Original class distribution — 500 non-diabetic (65.1%) vs 268 diabetic (34.9%) — confirms need for SMOTE

Figure 2: Feature Correlation Heatmap

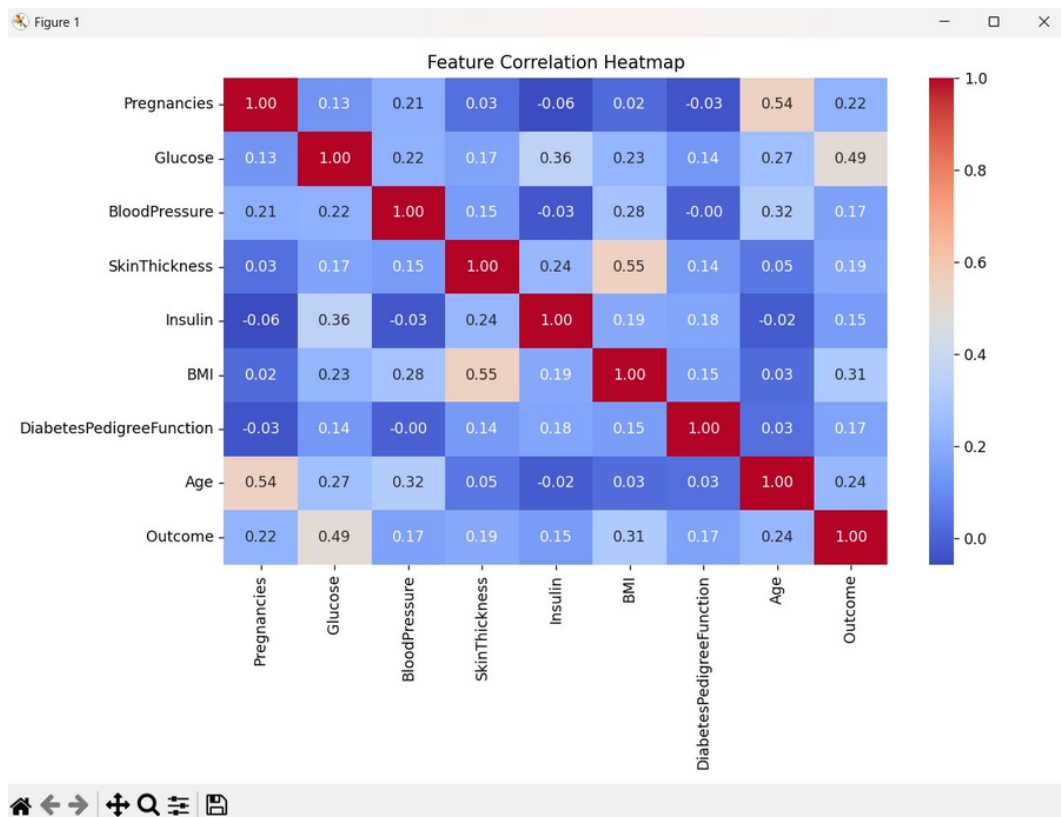


Figure 2: Pearson correlation — Glucose (0.49) has highest correlation with diabetes outcome

Figure 3: Feature Importance — Mutual Information

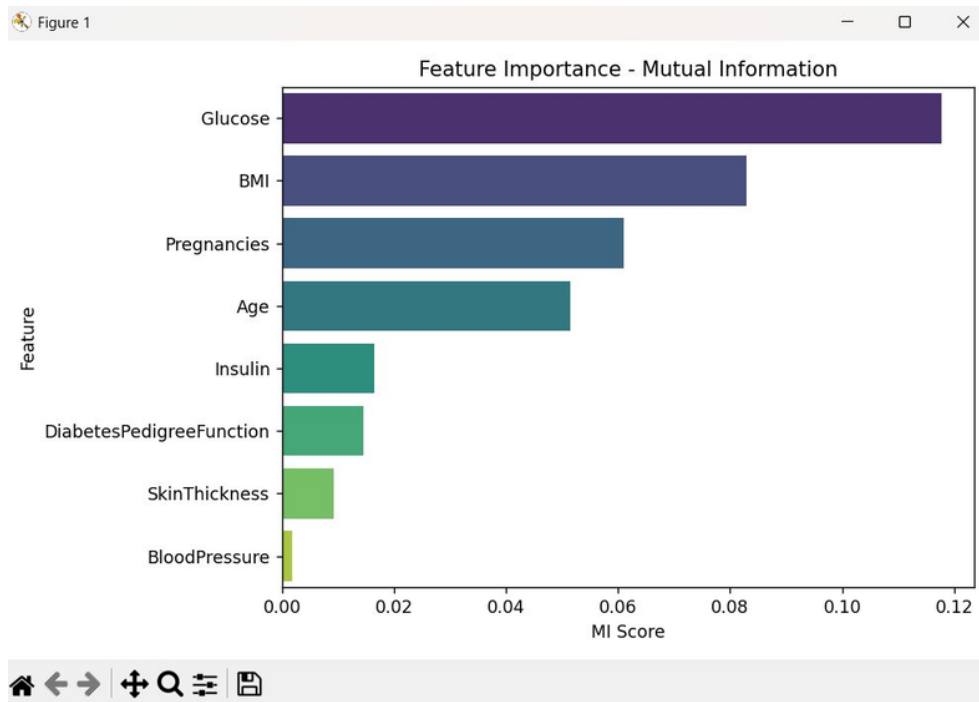


Figure 3: MI scores — Top 6 selected: Glucose, BMI, Pregnancies, Age, Insulin, DiabetesPedigreeFunction

Figure 4: Comparative Analysis — All Models

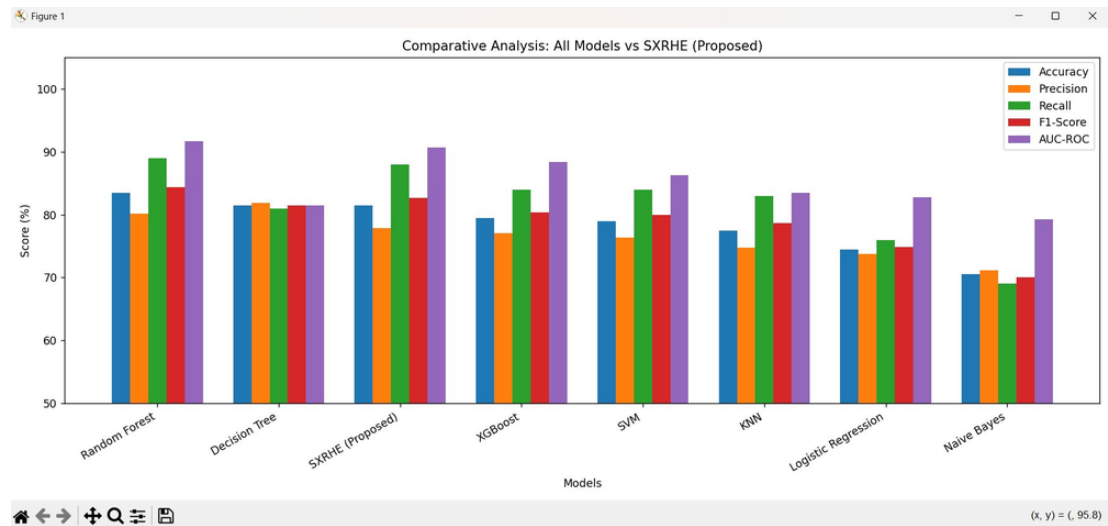


Figure 4: All 8 models comparison — SXRHE achieves highest AUC-ROC (90.61%)

Figure 5: Confusion Matrix — SXRHE

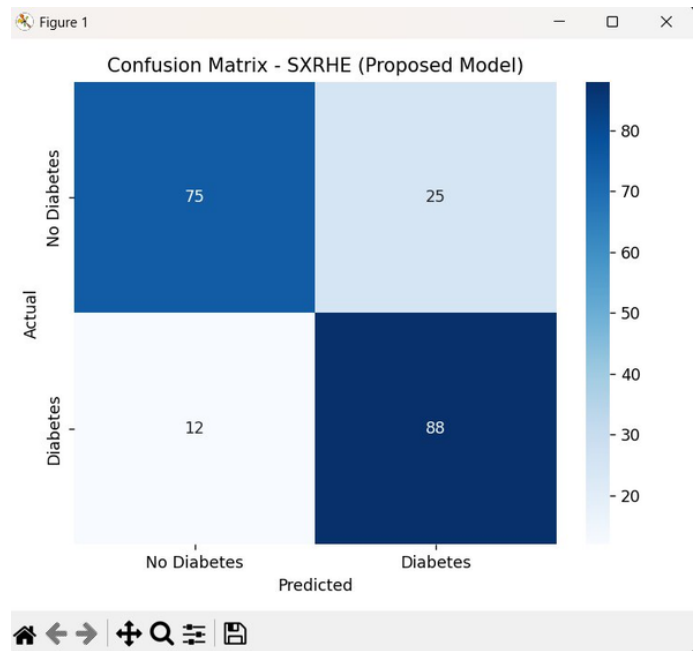


Figure 5: TP=88, TN=75, FP=25, FN=12 — high recall minimizes missed diabetic cases

Figure 6: ROC Curve Comparison

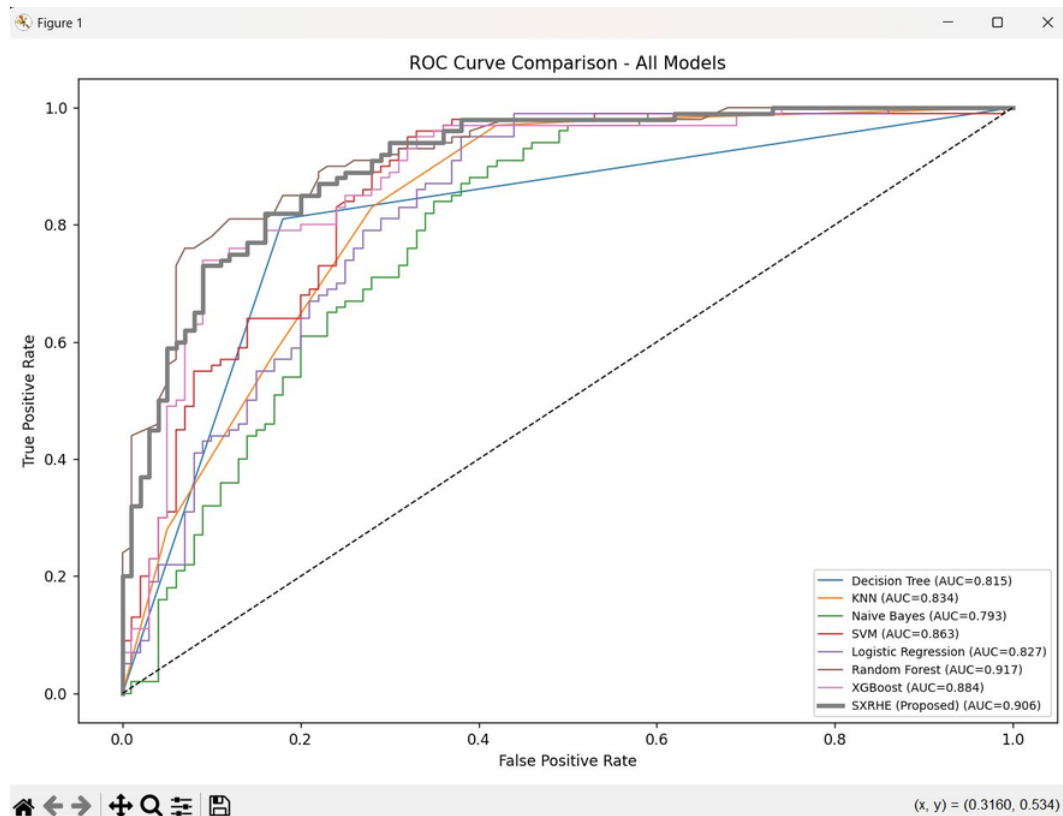


Figure 6: ROC curves — **SRHE (AUC=0.906)** in bold; demonstrates strong discriminative ability

Figure 7: 10-Fold Cross Validation Boxplot

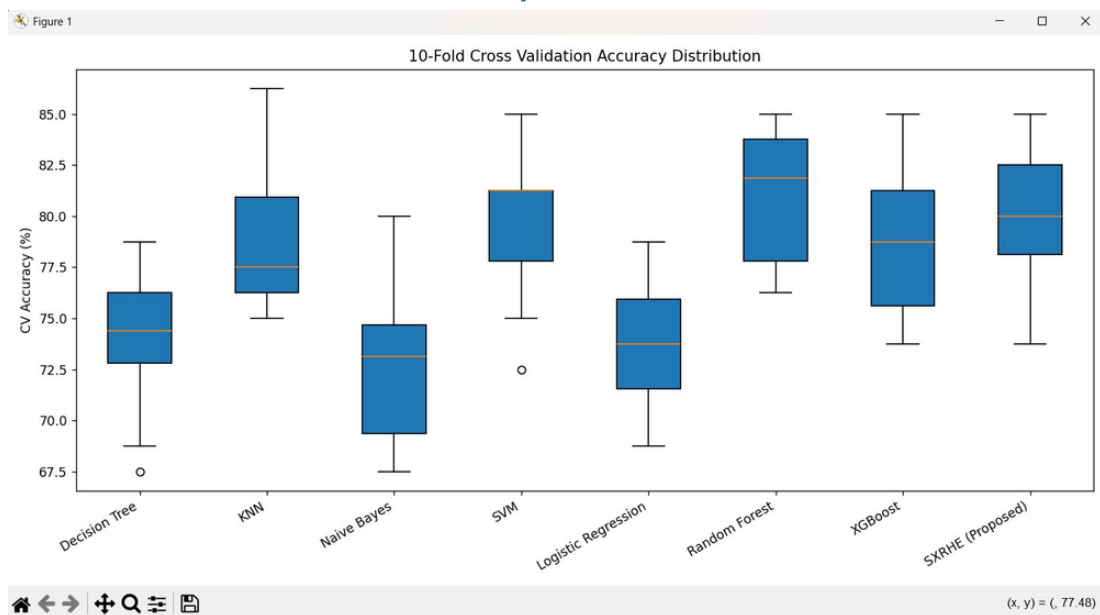


Figure 7: **SRHE** consistent ~80% accuracy with low variance across all 10 folds

Figure 8: SHAP Feature Importance Summary

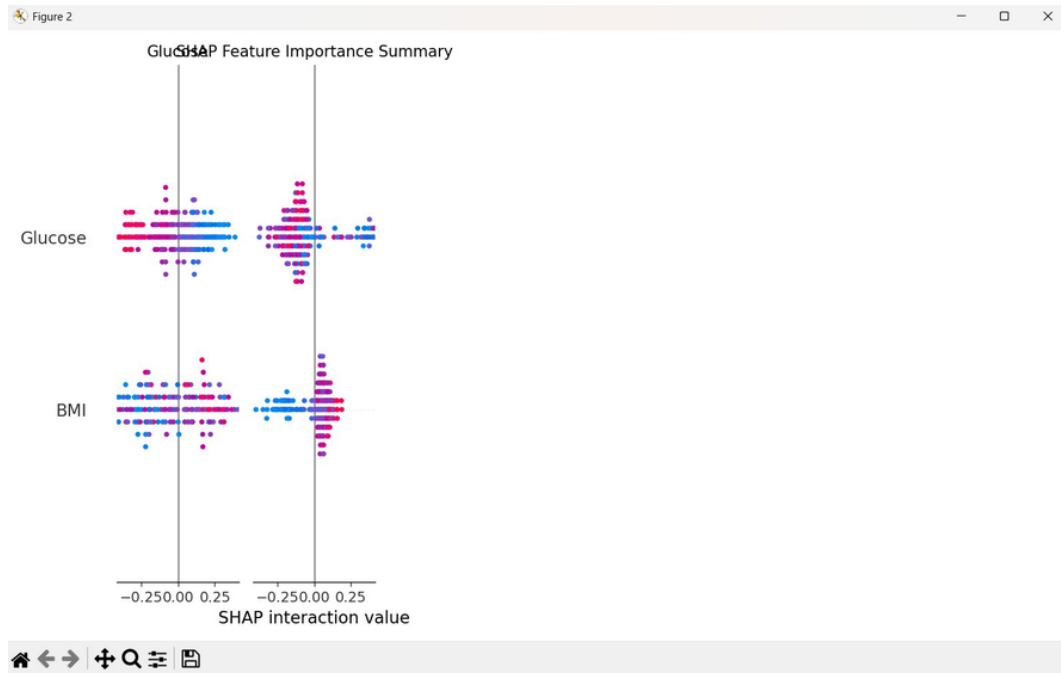


Figure 8: SHAP beeswarm — Glucose and BMI highest impact; red=high value, blue=low value

Figure 9: SHAP Mean Feature Importance

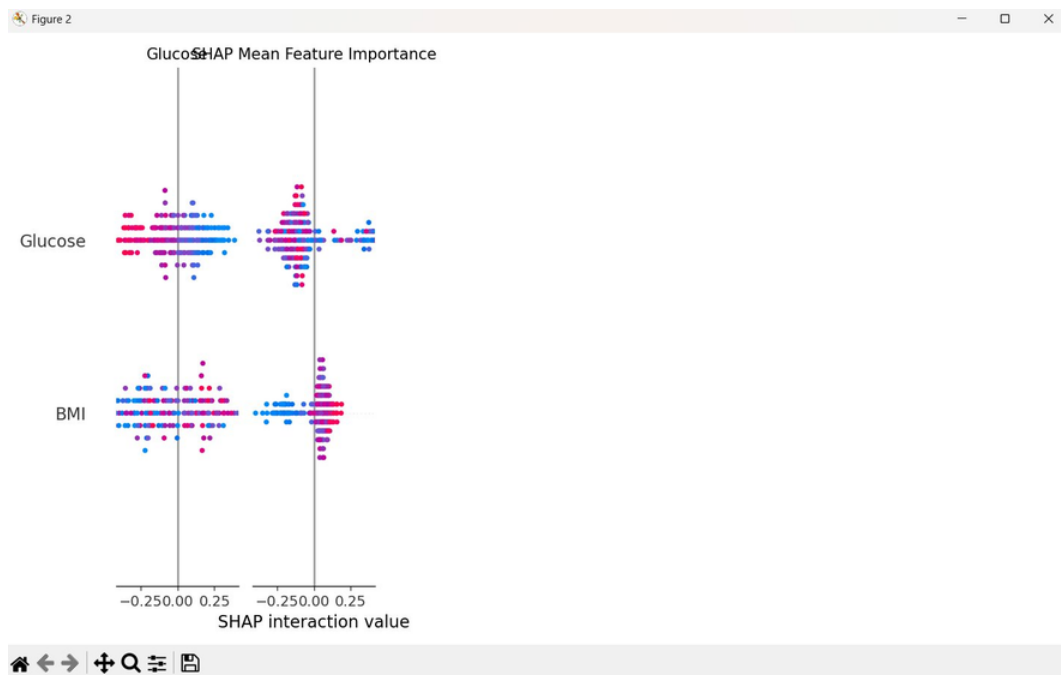


Figure 9: SHAP mean absolute values confirm Glucose as dominant predictor followed by BMI

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